

The accuracy of benefit-cost analysis for transport projects supported by the Asian Development Bank

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ABSTRACT

This paper analyses the accuracy of Benefit-Cost Analysis (BCA) of transport projects financed by the Asian Development Bank (ADB). *Ex post* evaluation of BCAs outside Europe and North America has been largely missing from the current literature. This study covers 59 roadway projects in developing countries funded by ADB and examines the accuracy of BCA results through four dimensions: frequency, magnitude, correlation, and tendency. We found that under an average project completion rate of 95.33%, the average construction cost overrun for 59 roadway projects is 10.71%, equivalent to USD 71.4 million. Grounded on 23 projects disclosing detailed economic analysis, we discovered a systematic tendency to understate both the present values of cost (18 out of 23 projects) and the present values of future economic benefits (13 out of 23 projects) in *ex ante* BCA. Furthermore, more than half of projects (25 out of 47) underestimated EIRR, and about 52.17% of them (12 out of 23) understated NPV. Since the underestimation of economic benefits is too small to counterbalance the underestimation of costs, the project EIRR is on average 5.4% lower than the initial expectation. Moreover, we discussed ADB's choice of the social discount rate, shadow price, and counterfactual base scenarios, which significantly influence the accuracy of BCA results and the reliability of decision-making grounded on BCA results. Lastly, the causes of cost overruns and benefits underestimations were analyzed.

1. Introduction

In *The Wealth of Nations*, Adam Smith stated that "The third and last duty of the sovereign or commonwealth, is that of erecting and maintaining those public institutions and those public works" (Smith, 1910). Transport infrastructure accounts for a significant portion of public spending and is critical in facilitating local and international trade and cultivating economic growth (Russ et al., 2018; Walters, 1967). Demonstrating the worthiness of investment in transport projects like road construction projects is a demanding and sophisticated job, and Benefit-Cost Analysis (BCA) is the core tool for this job (Lee Jr, 2000). As a quantitative analytical tool, BCA synthesizes a basket of quantifiable primary elements directly relevant to a prospective project and produces multiple expected performance indicators as outputs to assist decision-making (Joseph et al., 2020). Those primary elements are subdivided into cost items, such as cost of civil works, financing costs, project management, and consulting cost, as well as benefit items, including vehicle operating costs (VOC), safety benefits, and travel time saving (TTS) benefits for new and existing travelers. Performance

indicators output by BCA typically include the Benefit-Cost Ratio (BCR), Net Present Value (NPV), and Economic Internal Rate of Return (EIRR). Those primary elements and performance indicators may vary with project type and the requirements of project investors or stakeholders, but they are roughly consistent when evaluating comparable projects of similar kinds.

As demonstrated by ample literature regarding the *ex post* analysis for transport projects in the developed world, measuring the extent to which BCA's results are consistent before and after project completion delivers important value for future project evaluation and monitoring. However, limited attention has been paid to the BCA applied to transport projects in developing countries and its accuracy. This paper attempts to supplement the current literature by expanding the focus on *ex post* BCA of transport projects in the developing world. It does so by focusing on roadway projects funded by the Asian Development Bank (ADB), which possess both *ex ante* and *ex post* BCA that were prepared in a somewhat similar way.

Key research questions are.

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1. Which BCA primary elements (cost or benefit items) are most vulnerable to errors?
2. How much do those BCA primary elements and other project-specific characteristics contribute to the inaccuracy of the performance indicators output by BCA?
3. What is ADB's current practice in terms of the social discount rate, shadow price, and counterfactual base case?

This paper is organized as follows: section 2 groups the existing literature on actual project performance assessment into four categories, including project costs, travel demand forecast modelling, other comprehensive project benefits, and overall project performances, and details main findings under each category; section 3 and 4 state data source and statistical analytical methods adopted by this study; section 5 outlines results from four accuracy assessment dimensions, that being frequency, magnitude, correlation and tendency; section 6 discusses three methodological issues associated with the BCA approach adopted by ADB and the causes of cost overruns and benefits underestimations.

2. Literature review

Scholars care about the actual project performance after project delivery for large-scale transport projects. As documented by the existing literature, project cost evaluation and travel demand forecasts lie at the heart of the concern, with extensive research on how projects in developed countries behave in those two dimensions published. In recent years, a gradually growing literature on other comprehensive project benefits such as economies of agglomeration and impacts on land use and the overall project performances has been observed. The following subsections delineate the current research practice for each of the four topics.

2.1. Project costs

Ex post evaluation on project cost performance pays attention to how the ultimate out-turn cost deviates from the budgeted amount that was stated at various *ex ante* stages, which is more popularly known as the extent of cost overrun. Cost overrun is measured in both absolute monetary value and percent error manner, enabling to report its severity and detect the existence of any systematic tendency. In addition, statistical models (mainly regression models) are engaged to test and predict the correlation between cost overrun and factors like project-specific traits, temporal or geographical features, and external economic conditions. Although projects subjected to *ex post* cost evaluation spread all over the world, the detection, supervision, and research on this issue initiated far earlier and are much more extensive in the developed world than in developing regions. Specifically, past academic studies and research addressing transport project cost performance include but are not limited to those published by Andrić et al. (2019), Cantarelli et al. (2012), Flyvbjerg et al. (2004), Huo et al. (2018), Lee (2008), Love et al. (2016), Lundberg et al. (2011), Odeck (2004), Park and Papadopolou (2012), Sebastian (1990).

Those studies revealed the universality of transport project cost overruns and delved into the correlations between the magnitude of cost overruns and a series of variables representing project characteristics like transport project mode and the project size reflected by the planned capital investment. First, Flyvbjerg et al. (2004) observed the largest mean cost overrun of roughly 45% for rails, followed by 34% for fixed-links, and 20% for roadways based on 248 projects in Europe and North America. Similar tendencies that rails are more vulnerable to over-budgeting in comparison to roadways and fixed-links have been collaborated by (Andrić et al., 2019; Huo et al., 2018; Lee, 2008; Park and Papadopolou, 2012; Sebastian, 1990). In contrast, the Netherlands study by Cantarelli et al. (2012) reported the mean cost overruns were approximately 22% for fixed-links, 19% for roadways and 11% for rails. The authors explained that the rails sampled by their study were mainly

up-grades to existing heavy rails instead of new construction, leading to a narrower cost overrun than rails elsewhere. Second, Andrić et al. (2019), Flyvbjerg et al. (2004) statistically proved a positive correlation between the size of cost overruns and project size, considering that the planned investment volume rather than the out-turn is the suitable measure of project size. However, Cantarelli et al. (2012), Odeck (2004) yielded a negative correlation, accentuating that smaller projects have larger overruns. In addition, Huo et al. (2018), Park and Papadopolou (2012) did not capture any statistically significant correlations between cost overruns and project size were captured.

Further, grounded on the responses to the questionnaires distributed to the companies participated in the project plan and build, Park and Papadopolou (2012) insisted that the overwhelming weight of bidders' quotations in the contractor selection criteria and the type of construction contract awarded are identified as the major drivers of cost overrun for the 35 transport projects from a mix of developing and developed countries in Asia. By reviewing the contribution of the causes identified in project reports and ranking the frequencies that those causes were nominated, Andrić et al. (2019) concluded that the major sources of project cost overrun related to construction resources and works, alterations to project design, land acquisition, and the volatility of currency price.

Moreover, Odeck (2017) conducted a quantitative meta-analysis on 48 contemporary studies of transport projects, systematically testing how the reported project cost overrun (in percent error form) changes with specific characteristics at the study level (such as geographical location and publication year). Among the 48 studies, only 5 studies were performed in Asia (4 from India and 1 from South Korea), 3 in Africa (1 from Zambia, 1 from Tanzania, and 1 from African Development Bank), and 1 in the Middle East (Jordan), with the remaining 41 studies belonging to developed regions like Australia, North America and Europe. As the only meta-analysis, he summarized, statistically integrated and critically extracted all comparable research findings in this field grounded on a sample size that triumphs over any single study. He concluded that the frequency of cost overrun exceeds cost underrun, that the percent error cost overrun drops over time, that cost overrun stated by published works is significantly higher than those in unpublished works, that roads are less likely to have cost overrun than other modes (rails and urban transport), and that cost overrun is found to be more severe in developed regions (except for Australia) than in Asia, Africa, and the Middle East.

2.2. Travel demand forecast modelling

Ex post evaluation on the accuracy of demand forecast, which constitutes the basis for evaluating direct economic benefits like travel time saving (TTS) and vehicle operating cost (VOC) reduction, are ample. The accuracy is measured by how the envisaged demand differs from the observed demand, mainly reported in percent error. This difference is also commonly referred to as demand shortfalls. The statistical analysis on demand shortfalls is similar to cost performances. Nevertheless, the models engaged may differ since travel demand is time-series data, requiring consideration of actual traffic states at various points in time, spread across the observable project operation period.

Previous studies on this topic have covered transport projects in various modes (mainly road and rail) worldwide, but there is significantly more research in developed than developing countries. Those academic publications include but are not limited to US transit projects by Pickrell (1989), Voulgaris (2019 a,b), Australian toll roads by Li and Hensher (2010), roadways in Minnesota by Parthasarathi and Levinson (2010), toll and toll-free roads in Norway by Welde and Odeck (2011), toll roads across the world by Bain (2009), roads of all types in the US and Europe by Hoque et al. (2022), and transport projects in various modes across the globe by Cruz and Sarmiento (2019), Flyvbjerg et al. (2005), Hartgen (2013), Nicolaisen and Driscoll (2014).

The main findings related to demand forecast in developed countries

are: first, the overestimation of travel needs is more prevalent than the underestimation, and the extent to which the forecast travel needs are overestimated varies depending on the time, size, location, and type of transport projects dataset used. For example, the discrepancy between forecast and actual travel demand is more extensive for transit projects than for roads (Cruz and Sarmiento, 2019; Flyvbjerg et al., 2005).

Second, controversial findings of whether the size and direction of forecast errors vary over time exist. Flyvbjerg et al. (2005) argued that over the three-decade period they covered, the size of the demand forecasting errors for rail projects remains constant and that the accuracy of roadway demand forecasting even got worse. The authors stressed that the crux of the matter is not fixing the forecasting models but calibrating the underlying assumptions by engaging the lessons learned in a systematic and proactive manner. Cruz and Sarmiento (2019) also confirmed the failure to reduce forecast errors over time grounded on 21 studies established between 1978 and 2017. Nevertheless, Hoque et al. (2022) reported that based on 1291 roads in the US and Europe, the traffic estimations of more recent roads were closer to the actual traffic counts in comparison to roads opened earlier. The authors explained that modified forecasting modelings and more sufficient data may be attributable to this change. In addition, the size of forecasting error for the same transport project is expected to gradually decrease as the project operation time becomes longer because the ramp-up effects fade away (Bain, 2009; Cruz and Sarmiento, 2019; Li and Hensher, 2010). For the direction of error, the propensity for understating actual travel needs for roads opened before the early 2000s has turned out the opposite for roads opened more recently (Hoque et al., 2022). Overstating travel demands implies that the prospective transport facilities may reach capacity later than expected while understating are just the opposite. The authors believed that the Great Recession had contributed to such an apparent shift. In addition, the authors noticed an interplay between the direction of demand forecasting error and aggregate Vehicle Miles Travelled (VMT). VMT per capita grew in the two decades before 2000, during which a trend of underestimating road demands was observed. After 2000, VMT per capita decreased and then increased, and the estimated traffic demands on average exceeded the actual ones. Fuel price fluctuation and changes in economic conditions that impact VMT may also influence the changes in traffic demand forecasts.

Third, the inaccuracy of demand forecasts can be attributed to.

1. Defect modelling engendered by imprecise estimation on exogenous variables (like unemployment rate (Hoque et al., 2022), growth in car ownership and energy price (Jacoby and Minten, 2009; Nicolaisen and Driscoll, 2014), and the way that households access transport services (Fafchamps and Hill, 2005)) or on project-specific features (such as adjustment to project design (Nicolaisen and Driscoll, 2014) and delay of project delivery (Parthasarathi and Levinson, 2010; Voulgaris, 2019a b);
2. Scant awareness of the impact imposed by the bureaucratic environment (Flyvbjerg et al., 2005; Hartgen, 2013);
3. Bias caused by psychological restrictions like optimist bias (Bain, 2009; Nicolaisen and Driscoll, 2014); and
4. Insufficient consideration about or the intrinsic unpredictability of the impact of the project on land use and integral network arrangement (Nicolaisen and Driscoll, 2014; Parthasarathi and Levinson, 2010).

2.3. Other comprehensive project benefits and disbenefits

In spite of the concerns about direct benefits relying on demand forecast modelling mentioned, we observed that discussions on other comprehensive benefits generated by transport investments are gradually increasing, which comprise: 1) wider economic benefits like economies of agglomeration and spatial spillover effects, and 2) influence on land use which may be reflected by changes in transport accessibility

and in housing price.

As society transforms from agrarian to urban, the density of business entities increases by improving cost efficiency and transport connectivity, which ultimately converts to non-agricultural productivity benefits or economies of agglomeration (Graham, 2007; Legaspi et al., 2015; Wright et al., 1981). Spatial spillover effects accrue to transport infrastructure when its positive influence on local productivity and economy spill from the target region to surrounding areas (Munnell, 1992). Economies of agglomeration and spatial spillover effects induced by transport investment, which are vital components of wider economic benefits generated by transport projects, have been one of the most persistent topics of interest for western scholars, and many excellent empirical studies include but are not limited to Venables (2004)'s study in London, Legaspi et al. (2015)'s study on Sydney North West Rail Link, Graham (2007)'s work grounded in the UK, Kernohan and Rognlien (2011)'s study in New Zealand, Cohen and Paul (2004)'s research in the and Alam et al. (2022)'s work covering 40 transport projects funded by Asian Development Bank (ADB), Japan International Cooperation Agency, and World Bank.

However, in the developing world, adjacent regions and countries that should have been the beneficiaries of the positive spillover effects are likely to be replaced by oversea countries, especially when the pillar industries (like trading basic agricultural commodities and unprocessed natural resources) heavily depend on trading and logistics. For example, the sharp rise in transport infrastructure linking inland mines to coastal ports in Africa is primarily driven by facilitating bilateral international (hard) commodity trades (Bonfatti and Poelhekke, 2017). The countries or companies that do not have access to those particular transport networks are forced to confront somewhat higher trade costs, diminishing their competitive advantages and exacerbating their living environments (Bonfatti and Poelhekke, 2017).

In addition, the economies of agglomeration may lead to different industrialisation patterns and polarised environmental consequences in wealthy regions and in poor regions in the long run. The intensified expressway networks in China enable poor villages and towns to foster local economic growth by accommodating polluting industries that are rejected by and transferred from the wealthy regions (He et al., 2020). In contrast, after evacuating the environmental unfriendly industries, the rich areas undergo industrial modernisation without losing industrial supply (He et al., 2020).

Those benefits and disbenefits account for a great portion of the long-run economic values delivered by transport projects and cannot be fully captured by assessing the actual usage of transport facilities. Unfortunately, they are generally excluded from the current state-of-the-art BCA applied at the project level because of the lack of standardized methods of measurement and the presumed minor impact at the individual project level.

Transport infrastructure's effect on the land use pattern in adjacent and remote areas and housing prices is another topical issue. Although such an effect is normally analyzed in feasibility studies carried out at different project planning stages, it has typically only been performed qualitatively. The change in accessibility caused by a transport project is increasingly acknowledged as the kernel of quantitatively capturing such an effect. Accessibility is measured as the ease of reaching valued destinations (COTAM, 2020; Hansen, 1959), encompassing not only how fast people and goods can travel around (mobility) but also the spatial distribution of opportunities (Levinson and King, 2019). The dynamic interplay between the level of change in accessibility and that in housing price provides an opportunity to assess how much economic benefits are materialised as a result of the construction of a transport project (Adriano Borges Costa and Zheng, 2021; Beenstock et al., 2016; Mohring, 1961; Wang and Levinson, 2022). Substantial efforts have been carried out by scholars in developed countries, such as the work by Riekkinen et al. (2015) in Finland, by Dewees (1976) in Canada, by Agostini and Palmucci (2008) in Chile, and by Hess and Almeida (2007) in the US. However, other comprehensive project benefits are popular

mainly in research instead of in practice, restricting the presence of comprehensive large-N studies comparing *ex ante* estimation and *ex post* observation.

2.4. Overall project performance

Overall project performance, focusing not just on individual elements of BCA but also on integral performance indicators like NPV, EIRR, or BCR, is the last area of research interest, which is also our primary concern. It is notable that the literature covered here does not overlap with those listed in previous sections.

Participants in this research field include government agencies and academic researchers. Nicolaisen and Driscoll (2016) summarized 8 existing *ex post* evaluation programs hosted mainly by national government agencies in OECD countries and conducted further analysis on the programs in the UK (Post Opening Project Evaluation scheme) and Norway. The authors ruled out *ex post* evaluation programs carried out by multilateral development organizations such as the World Bank because inconsistent performance measures are adopted by different organizations and the ability to verify data is restricted. The authors' concerns about the quality and efficiency of those programs are addressed by developing a typology based on characteristics of each program, such as the type and number of transport projects subject to *ex post* evaluation, the time point of *ex post* analysis and etc. For instance, for projects operated by Highways England, the authors identified a proclivity for understating costs, but that does not appear in the context of benefits evaluation. In the case of Norwegian roads, the likelihood of cost overrun is the same as that of cost underrun, and more than anticipated benefits are materialised. In addition, Nicolaisen and Driscoll (2016) also underlined that data unavailability, difficulties in consistently determining counterfactual scenarios in *ex ante* and *ex post* stages, and incomparability among projects in different regions due to the adoption of highly customized *ex post* evaluation approaches restrict research on *ex post* project evaluation.

In comparison to the aforementioned three research areas, the number of academic publications delving into this topic is relatively sparse (Boardman et al., 1994; Ika and Feeny, 2022; Nicolaisen and Driscoll, 2016; Odeck and Kjerkevit, 2019). To the best of our knowledge, studies at large-N level (cover at least 10 projects) typically target projects in OECD countries like France (Meunier and Welde, 2017), Norway (Odeck and Kjerkevit, 2019; Welde and Volden, 2018) and the Czech Republic (Halánek et al., 2021) or projects financially supported by European Commission (Florio and Vignetti, 2013; Jong et al., 2019; Kelly et al., 2015).

By providing empirical statistical evidence, those studies are conducive to formulating a holistic view of how project outcomes shift from initial anticipation based on statistical evidence, thereby identifying defects in the evaluation approach applied during planning (Kelly et al., 2015). For example, in the study on Norwegian roads, Odeck and Kjerkevit (2019) confirmed the existence of systematic underestimation of benefits and added that *ex ante* project NPV were understated by roughly 50% because the extent to which costs were understated is too petty to counterbalance the underestimation of benefits. The authors subsequently highlighted the causes of benefit underestimation are the improper base for demand forecast (relying on regional forecast instead of the project-particular one) and the prolonged interval between official project decision-making and the completion of *ex ante* BCA.

Research at the case study level is prone to select high-profile transport projects in developed countries, which consume massive investment and is expected to persistently influence the local society decades into the future, like the Coquihalla Highway in Canada (Boardman et al., 1994), the Channel Tunnel linking the UK and France (Anguera, 2006), the Oresund Bridge in Denmark (Knudsen and Rich, 2013), the Bolu Mountain Tunnel in Turkey (Kocabas and Kopurlu, 2010) and the Stockholm Metro in Sweden (Börjesson et al., 2014).

From the above literature, the imbalanced input for research on it

between developed countries (mainly European countries) and developing countries is prominent, calling for further research evidence of the latter. In addition, a few questions about the fundamentals of evaluation methodology, including the determination of the shadow price, the choice of discount rate, and the identification of counterfactual "no-build" or "do-minimum" base case upon which incremental costs and benefits are assessed, have been repeatedly raised and discussed. A discussion about how the developed countries address those problems during project evaluation and how the current practice might be optimized is needed.

3. Data

There are numerous International Financial Institutions (IFIs) providing funding and knowledge support to aid developing countries either worldwide, such as the World Bank, or regionally, such as the Asian Development Bank (ADB), the Japan International Cooperation Agency (JICA), and the African Development Bank. The *ex ante* project appraisal methods and procedures, project decision flows, the process and criteria of funding disbursement, and the *ex post* project evaluation schemes differ across IFIs, resulting in an issue regarding the comparability of the source and derivation of project data values. As a result, it was determined to stick with one IFI - the ADB to reinforce data homogeneity and sampling robustness.

To avoid the heterogeneity caused by differing modes of transport projects, this study targets 59 roadways funded by ADB and erected in developing countries in Asia and Oceania. Those roads were approved between 1998 and 2010, constructed between 2004 and 2018, and evaluated between 2004 and 2020. The 59 projects are documented in "ADB's Success Rates Database 2010–2018", which is a dataset produced by ADB and ADB Independent Evaluation Department (IED). The original dataset includes 72 roadway projects, but we finally retained 59 projects (82%) for *ex post* cost analysis and 23 projects for *ex post* economic analysis (32%). The selection criteria include:

1. Keep complete record of *ex ante* and *ex post* project cost breakdown
2. Disclose both *ex ante* and *ex post* project Economic Internal Rate of Return (EIRR).
3. Disclose detailed net benefit streams throughout the anticipated project life.
4. With reliable records of historical national economic indices like Consumer Price Index and exchange rate to USD.

The average inflation-adjusted project cost estimated in the appraisal stage is \$US 563 million (with a standard deviation of 574), and the average inflation-adjusted actual project cost is \$US 623 million (with a standard deviation of 721). Among the 23 projects, it is worth noting that the base year chosen to calculate NPV is inconsistent. On the one hand, among different projects, the construction beginning year, project opening year, or even a few years post-to project opening were observed to be used as the base year. On the other hand, for the same project, the selected base year changed between *ex ante* and *ex post* evaluation stages. Since the choice of the base year has a great influence on the final NPV, we designated the construction beginning year as the base year and recalculate the NPV for all 23 projects.

According to Asian Development Bank (2016 a), ADB provides various public sector financial assistance including loans (Ordinary Capital Resource (OCR) Loan), grants (Asian Development Fund), and technical assistance for its Developing Member Countries (DMCs) to support regional development and economic growth. From the perspective of ADB, sovereign credit rating and the level of per capita income are fundamental factors affecting the type of financial support granted. Roadway projects sampled by this study were those granted the OCR loan.

For each completed sovereign operation project in which the ADB invests in part or in full, ADB requires a two-step *ex post* evaluation

process. First, it mandates a self-assessment under the supervision of ADB's regional department, producing the project completion report (PCR) (Asian Development Bank, 2016 a). A PCR provides the results of an assessment on the performance of a project through four critical dimensions: relevance, effectiveness, efficiency, and sustainability (Asian Development Bank, 2016).

- 'Relevance' delves into the level of strategic alignments between the intended project and the sovereign's development priorities, the conformity of the expected project outcomes by ADB's rules and strategies, and the fitness of the project design for delivering the expected outcomes.
- 'Effectiveness' assesses the extent to which the expected project outputs, represented by the project baselines and planned targets, are substantially realised when the project is completed and detects the existence of unexpected outcomes that corrode the project's value.
- 'Efficiency' focuses on how successfully the resources are utilized to deliver the planned outcomes, which is central to our concerns. It is measured by EIRR, NPV, or BCR, which are recalculated and compared against the *ex ante* ones to decide if society will benefit from the net economic return generated by the project. ADB highlights that if the methods of recalculating EIRR in PCR differ substantially from that applied in *ex ante* stages, such as the alteration to underlying assumptions, parameters, discount rates, and the expected project lifespan, further clarification is warranted. For the projects sampled in this paper, we only noticed minor differences between the expected project lifespan stated in *ex ante* and *ex post* evaluation stages, but no explanations were attached. And some of the benefits which had been identified in project appraisal were eliminated in PCR. For those benefits and costs identified and extant in both stages, we did not observe any further clarification on substantial changes in the calculation methods, the parameters, and assumptions.
- 'Sustainability' considers the chance that project outcomes and outputs will persist over the project's foreseeable lifespan, which covers financial (the income-generating ability against the upfront capital costs), institutional, and environmental sustainability aspects.

ADB stipulates that a PCR is expected to circulate across the ADB's board and be available on the website within 1–2 years of project completion (the official closure of the project loan account).

Second, this PCR will be subsequently assessed and justified by ADB's IED Asian Development Bank (2016 a). The IED, which does not participate in the preparation of PCR conducts an evaluation as either: a Project Validation Report (PVR) or a Project Performance Evaluation Report (PPER). Those two forms of independent evaluation differ in the breadth and depth of assessment, reflected by the duration of assessment, the manner of evidence collection, and finally evidence and analytical resources relied on. Asian Development Bank (2019) specifies that PVR is a quick desk review type of assessment which generally takes 2–3 weeks and relies heavily on information reported in previous documentation like PCR. Given the rapidity of the PVR, roughly 80% of completed projects are independently assessed this way. The PPER is a field-based sophisticated assessment which typically consumes 3 months or so to prepare and adopt field visits, surveys and interviews to collect information and examine the validity of the PCR. Considering such a heavy investment of manpower and time, about 20% of completed projects go through PPER. Given the limited information available in the initially extracted dataset, we scoured PCRs for all 59 projects (and PPERs and PVRs where available) to collect additional information.

Although we tried the best we could to collect all available information about those projects in the developing countries, the lack of national open data portals recording the actual traffic states and overhead costs post to project opening restricts independent cross-validations and the traceability of the actual usage of those public

facilities in the following years. It is noteworthy that in comparison to the retrospective project analysis in developed countries where the risks of projects being distorted by vested interests and potential political biases are more effectively identified, disclosed and hedged by using transparent and objective actual operation data, relying solely on the *ex post* project analysis produced by ADB is somewhat uncertain.

4. Methodology

As a measure of the inconsistency of BCA results between *ex ante* and *ex post* project stage, the error is defined as the numerical discrepancy between the two stages. Various costs, economic benefits, and expected performance indicators output (mainly NPV and EIRR) are subject to the assessment of errors. We proposed four dimensions from which the errors can be assessed.

First, *frequency of error* is defined as the frequency of observed discrepancy between *ex ante* and *ex post* BCA. It is computed by the number of projects reporting discrepancy divided by the total number of projects collected. This dimension identifies the likelihood that each element of BCA are vulnerable to inaccurate estimation. Second, *magnitude of error* describes the magnitude to which the *ex post* results deviate from the corresponding *ex ante* estimation. It is measured with two patterns: errors in absolute value (E_{abs}) and errors in percent form (E_{per}). The former is calculated by equation (1), meaning that the absolute error of project i equals to its observed value in *ex post* stage (O_i) minus its forecast value in *ex ante* stage (F_i). And the arithmetic mean ($\overline{E_{abs}}$) is calculated by the summation of E_{abs_i} over the total number of projects (n). As shown by equation (2), E_{per_i} is calculated by the Mean Absolute Percent Error, and the arithmetic mean $\overline{E_{per_i}}$ is dividing the summation of E_{per_i} by the total number of projects (n). This dimension provides summary statistics showing the severity of estimation inaccuracy, and its result will be used in the following dimensions.

$$E_{abs_i} = |O_i - F_i| \quad \text{and} \quad \overline{E_{abs}} = \frac{1}{n} \sum_i^n E_{abs_i} \quad (1)$$

$$E_{per_i} = \frac{O_i - F_i}{F_i} \quad \text{and} \quad \overline{E_{per}} = \frac{1}{n} \sum_i^n E_{per_i} \quad (2)$$

Third, *correlations of error* aims at capturing and quantifying the statistical interaction among project characteristics, changes of economic condition, and results output by BCA. Multivariate linear regression is engaged in this dimension, and variables fed into regression analysis are shown as per Table 1. Fourth, *tendency of error* identifies if certain types of errors present a temporal or regional tendency. It is measured primarily by engaging continuous time variable Y2000 (variable 9) and interaction terms (variable 11) in Table 1 via regression analysis.

5. ADB results

5.1. Frequency of error

Results for the first assessment dimension on initial project construction cost and economic analysis are shown in Table 2 and Table 3 respectively.

In Table 2, project contingency, cost of civil works, and land acquisition cost rank at the top three in terms of the likelihood of suffering from *ex ante* underestimation. Project contingency, occupying about 10% of total costs, is reserved to cover costs incurred due to any unforeseeable events. Its 0.05% weight at *ex post* stage shows that it almost has been used up during project implementation. This is also confirmed by the fact that 98.08% projects have an ending balance of contingency lower than the beginning balance. As the largest single cost component, cost of civil work accounts for 70.74% (*ex ante*) and 83.35% (*ex post*) of the total initial project cost. We see that roughly 88% (52 out of 59

Table 1
Overview of variables in regression analysis.

ID	Name	Type ^a	Description	Differs in <i>ex-ante</i> and <i>ex-post</i> ?
1	ADB Funding %	C	The proportion of funding disbursed by ADB in total project costs	✓
2	Implementation Period	C	The number of years between the start and end of project construction	
3	Initial Construction	D	1 for a new construction project, otherwise 0	
4	Subsequent Construction	D	1 for a project that aims to retrofit, rehabilitate, widen or extend the existing roads, otherwise 0	
5	Access Road	D	1 for an access road, otherwise 0	
6	Limited-Access Road	D	1 for a limited-access road, otherwise 0	
7	Access and Limited-Access Road	D	1 for a project consisting of both access and limited-access roads, otherwise 0	
8	Length	C	The centerline length (kilometer) of a road.	
9	Y2000	C	The number of years between a project's opening year and the base year 2000.	
10	Project Life	C	The proposed project life of a road.	✓
11	Location · Length	IT	An interaction term combining a road's geographic location and its length (Locations include South Asia, East Asia, Central West Asia, Oceania, and Southeast Asia (base group))	✓
12	PVC	C	The accumulated present value of the total costs for a project, including capital costs and maintenance & operation costs	✓
13	PVB	C	The accumulated present value of the total benefits for a project, including VOC savings, TTS, and other benefits	✓
14	NPV	C	The difference between a project's PVC and PVB.	✓
15	EIRR	C	The discount rate at which a project is break even.	✓
16	ΔCPI	C	The variation of local CPI during the period of project construction.	

^a C refers to a continuous variable, D refers to a dummy variable, and IT refers to an interaction term.

projects) of projects ended up with a higher than expected civil work cost. However, only three-fifths of the 59 projects experienced total project cost overrun, implying that contingency reserve effectively functioned as a buffer and mitigated the risk of cost overrun. Although land acquisition cost comprises a small share of total cost, there is a 56% possibility (23 out of the 41 projects that incurred this cost) of exceeding initial expectation.

As shown in Table 3, the number of projects disclosing detailed *ex ante* and *ex post* economic analysis is less than that providing detailed cost breakdown. A total of 47 projects reported *ex ante* and *ex post* EIRR, while only 23 road projects disclosed detailed economic analysis. For those 23 projects, 18 of them underestimated the present value of cost in *ex ante* analysis. Although capital cost occupies the largest weight in terms of the total cost, the likelihood of underestimating capital cost and

Table 2
Frequency of errors in initial project construction cost.

	n ^a			<i>ex ante</i> Stage		<i>ex post</i> Stage	
	Total	F<O ^b	F>O	W% ^c	Std ^d	W%	Std
Civil Works	59	52	7	70.74%	10.37	83.35%	9.88
Land Acquisition	41	23	18	3.02%	4.44	3.58%	5.80
Equipment	47	13	34	2.51%	4.57	1.51%	3.26
Consulting Service	54	30	24	4.46%	4.36	4.50%	4.50
Project Management	32	14	18	1.26%	2.18	1.16%	2.45
Interest Expense	54	17	37	4.07%	2.78	3.41%	3.75
Contingency	52	51	1	10.58%	5.60	0.05%	0.35
Other Costs	39	15	24	3.36%	4.61	2.45%	4.12
Total Cost	59	36	23	100%		100%	

^a n stands for the number of projects.

^b F refers to the forecast value at *ex ante* stage, O refers to the observed value at *ex post* stage.

^c W% represents the average weight of the cost item out of the total cost.

^d Std denotes the standard deviation of W%.

Table 3
Frequency of errors in project economic analysis.

	n			<i>ex ante</i> Stage		<i>ex post</i> Stage	
	Total	F<O	F>O	W%	Std	W%	Std
Capital Cost	23	16	7	95.63%	4.08	98.01%	17.61
O&M Cost	22	14	8	4.37%	4.08	1.99%	17.61
Total PVC	23	18	5	100%		100%	
Vehicle Operating Cost Saving	23	11	12	73.59%	21.22	68.57%	16.22
Travel Time Saving	21	19	2	15.09%	20.10	25.83%	13.85
Other Benefits	16	7	9	11.32%	12.27	5.60%	7.31
Total PVB	23	13	10	100%		100%	
EIRR	47	25	22				
NPV	23	12	11				

operation & maintenance cost are roughly equal. We saw that 13 projects had underestimated the total present value of future economic benefits. Among the three types of benefit, Vehicle Operating Costs (VOC) savings shares the largest portion of total benefit and is almost equally likely to be over- and underestimated. The portion occupied by Travel Time Savings (TTS) is much lower than by VOC, and 90% of those projects (19 out of 21) reported underestimation of TTS in *ex ante* stage. As for economic effectiveness measures, 25 projects (53.19% of 47 projects) underestimated EIRR and 12 projects (52.17% of 23 projects) underestimated NPV.

5.2. Magnitude of error

As shown by Table 5, the average overrun of the total cost for 59 projects is 10.71%, equivalent to USD 71.4 million. One may question whether project costs exceeded because of project scope expansion. Table 4 shows that on average only 95.33% of the originally planned projects were actually delivered.

The largest percent cost overrun is identified in civil works, followed by land acquisition cost and consulting services cost. In terms of cost underrun which is indicated by a negative E_{per} , project contingency has a mean percent error of 99.69%, again indicating that almost all contingency reserves have been used up during project implementation. The – presented in the penultimate column in Table 5 can be interpreted as either a zero ending balance or a cost item that hasn't incurred.

In terms of the assessment of economic analysis (as shown by

Table 4

Project completion percent.

Roads	n	Average Project Completion % ^a	Std
Access road ^b	16	98.15%	26.46
Limited-access road ^c	16	92.93%	14.87
Limited-access road and Access road	27	95.07%	22.77
Initial Construction	28	98.20%	19.45
Rehabilitation	31	93.62%	23.14
Total	59	95.33%	21.78

^a The project completion rate is calculated by the actual road length completed at the time of *ex post* evaluation divided by its planned length.

^b Access roads refer to roadways other than limited access roads.

^c Limited-access roads refer to motorways which can only be accessed from authorised entrances.

Table 6), the volatility of the overall results is far larger than the previous Table 5. A smaller sample size including only 23 projects may help explain this. In terms of total PVC, capital costs were underestimated by 45.74%, and regular maintenance & operating costs are 47.90% higher than planned. As for economic benefits, VOC savings were underestimated by 10.42% on average, but the average underestimation of TTS and other benefits are as large as 1313% and 249% respectively. This is because for many projects, TTS and other benefits were either not considered at all or not fully accounted for (eg. only consider TTS savings for freight transport but not for passenger transport) as potential project benefits at planning stage, but were re-considered at *ex post* stage. The — presented in the penultimate column (in Table 6) for other benefits shows that they have not been recognized as expected, at least not under the same category as planned, in the *ex post* evaluation stage. Subject to data unavailability, other benefits, such as savings on traffic crashes, poverty reduction and the boost to agriculture production via improved market access, are occasionally reflected as an added proportion to VOC savings or TTS (like scale TTS up by 10%), indicating

variation of the types of benefit considered in *ex ante* and *ex post* evaluation. Since the underestimation of economic benefit is too small to offset the underestimation of project cost, the overall EIRR is overestimated by 5.4%.

5.3. Correlation and tendency of errors

Models 1 & 2 in Table 7 summarise the regression results of total project cost and multiple independent variables. First, the amount of funding injected by ADB and project total length show statistically significant impacts on total project cost. Projects with heavier capital investment are expected to deliver longer roads. Notably, the length variable appears to be a more statistically significant estimator for project costs in the *ex ante* stages than in the *ex post* stage. As a key driver of project variable costs, length turned out to be less important than it had been stated in the planning stage, implying the existence of other influential fixed cost drivers independent of length which our model has not captured. The negative coefficient of ADB funding % indicates that the project earmarks from ADB may be limited, resulting in a relatively smaller proportion of ADB's funds for more expensive projects. Then, initial construction is more expensive than subsequent construction. Compared to projects containing both limited access and access roads, constructing either road type appears to be cheaper. This is plausible since projects containing both road types are likely to be larger. Moreover, our sample shows no statistically significant correlation between total project cost and local inflation. As suggested by the adjusted R^2 and the significance level for each statistically significant variable, model 1 outperforms model 2 in terms of the overall model fitness. The variables identified and included in our models are reliable when projecting total project costs in planning stages, but factors or uncertainties not captured by our model influence the actual out-turn cost in reality.

Models 3 & 4 describe the regression results of percent cost overrun (E_{per}) and multiple independent variables. First, cost overrun is independent of project size (represented by total cost), but it is positively

Table 5

Magnitude of errors in initial project construction cost.

	E_{abs} (in \$mil) ^a	Std	Lower ^b	Upper ^c	$E_{per}(\%)$ ^d	Std	Lower	Upper
Civil works	97.35	199.13	−70.30	1238.90	31.16%	32.22	−38.80%	132.91%
Land Acquisition	14.23	45.08	−25.60	258.20	25.13%	99.26	−13	378.18%
Equipment	−3.39	11.96	−73.50	15.90	3.50%	247.24	— ^e	1605.00%
Consulting Service	0.45	8.05	−40.30	29.00	15.69%	74.79	—	453.13%
Project Management	0.03	3.90	−20.92	12.89	−0.08%	85.82	—	304.00%
Interest Expense	6.11	35.62	−49.51	179.80	−14.00%	80.56	—	471.92%
Contingency	−41.15	56.10	−260.60	0.00	−99.69%	2.22	—	−84.00%
Other Cost	−2.24	8.85	−42.72	13.80	2.39%	112.51	—	422.22%
Total Cost	71.40	194.84	−142.81	1025.60	10.71%	26.24	−46.15%	99.26%

^a E_{abs} expressed in \$US million and calculated as per equation (1).

^b Lower means the lower value for E_{abs} or E_{per} .

^c Upper means the highest value for E_{abs} or E_{per} .

^d E_{per} as per equation (2).

^e “—” shows that the end balance of the cost item is zero.

Table 6

Magnitude of errors in project economic analysis.

	E_{abs} (in \$mil) ^a	Std	Lower	Upper	$E_{per}(\%)$	Std	Lower	Upper
Capital Cost	95.69	196.41	−169.39	764.68	45.74%	1.32	−75.30%	630.16%
O&M Cost	6.87	33.37	−37.40	131.84	47.90%	2.06	−471.74%	445.01%
Total PVC	102.55	196.76	−199.13	727.28	45.32%	1.31	−86.11%	623.19%
VOC Savings	−102.11	272.06	−774.38	408.41	10.42%	0.66	−78.51%	185.18%
TTS	115.23	261.06	−93.50	1079.04	1313.06%	51.90	−65.28%	23926.08%
Other Benefits	−36.92	99.08	−233.59	240.87	249.48%	11.43	—	4523.26%
Total PVB	−23.79	318.69	−1022.54	482.51	28.49%	0.74	−71.10%	232.68%
EIRR (N = 47)	−0.02	0.06	−0.18	0.08	−5.40%	0.28	−65.32%	50.20%
NPV (N = 23)	−126.35	339.47	−1406.54	175.61	45.66%	1.71	−197.41%	553.70%

^a E_{abs} expressed in \$US million except for EIRR, which is in %.

Table 7

Linear Regression models addressing the correlation between project-specific features (independent variables) and key performance indicators (dependent variables).

Dependent Variables	Project Cost		E_{per}		EIRR		<i>ex post</i> NPV	
	<i>ex ante</i>		<i>ex post</i>		<i>ex ante</i>		<i>ex post</i>	
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Constant	919.18*** ^c	843.40	−0.23	−0.20	0.21**	0.23**	1670.83**	208.97*
<i>ex ante</i> Total Cost			6.64E-05		−3.98E-05			
<i>ex post</i> Total Cost				1.07E-04		−4.31E-05*		
<i>ex post</i> PVC ^a							−0.05	
<i>ex post</i> PVB ^b								0.18
Implementation Period	−3.61	32.30	0.04	0.04	7.97E-03	0.01	−108.15*	−110.21**
Y2000	−1.96	−3.37	2.14E-03	3.99E-03	−1.49E-03	−1.62E-03	−9.68	−5.43
<i>ex ante</i> ADB Funding %	−774.53**		−0.05		−0.12*			
<i>ex post</i> ADB Funding %		−781.49**		−0.19		−0.03	−451.59	−112.09
Initial Construction	316.11**	385.17*	−0.08	−0.11	−0.05*	−0.06*		
Access Road	−363.51**	−322.35	0.04	0.09	0.14***	0.04	−381.52	−300.32
Limited Access Road	−126.06	−231.45	0.15	0.16*	0.03	−3.81E-03	2.81	48.34
<i>ex ante</i> Length	0.20***							
<i>ex post</i> Length		0.10*					1.50***	1.18**
ΔCPI	−194.77	−309.50						
<i>ex post</i> Project Life							−33.14	−25.79
<i>ex ante</i> East Asia · Length			7.03E-04		−7.35E-06			
<i>ex ante</i> South Asia · Length			−3.21E-05		2.05E-04*			
<i>ex ante</i> Central West Asia · Length			−5.22E-04*		3.37E-04*			
<i>ex ante</i> Oceania · Length			3.00E-03		−2.55E-03***			
<i>ex post</i> South Asia · Length				5.79E-04		4.42E-06		
<i>ex post</i> East Asia · Length				−2.28E-05		7.42E-05		
<i>ex post</i> Central West Asia · Length				−4.74E-04		1.52E-05		
<i>ex post</i> Oceania · Length				2.45E-03		−1.70E-03		
N	59	59	59	59	47	47	23	23
Adj. R^2 ^b	0.58	0.50	0.26	0.30	0.55	0.42	0.31	0.41
P-value	6.98E-09	5.51E-07	6.70E-03	2.40E-03	1.53E-05	8.34E-04	0.038	0.011

^a *ex post* PVC is the reevaluated present value of project cost, including capital costs and routine operation & maintenance costs.^b *ex post* PVB is the reevaluated present value of project benefits, including VOC savings, TTS and other benefits.^c *, **, *** indicate significance at the 90%, 95%, 99% and 99.9% level, respectively.

correlated with the project implementation period. Limited access roads have cost overrun greater than projects containing limited access and access roads. For the latter, the limited access road component generally consumes the majority of the project cost. As a result, when project budgets fall short, construction of the access road component may be postponed to ensure the completion of the limited access road. In other words, the budget for access roads functions as an extra funding reserve for the limited access road. In addition, the correlation between cost overruns and road length planned at *ex ante* stage shows geographic differences. In contrast with extending 10 km of road in Southeast Asia, doing it in East Asia would induce an extra 0.7% of percent cost overrun.

Models 5 & 6 in Table 7 show the regression result between project EIRR and multiple independent variables. First, project EIRR negatively correlates with total project cost in both *ex ante* and *ex post*. Then EIRR for initial construction is lower than that for subsequent construction. EIRR for access roads is higher than a project containing both limited access roads and access roads. Again, geographic differences are observed. EIRRs delivered by projects in South Asia and Central West Asia is higher than that of projects in Southeast Asia. Constructing 10 km of roads in South Asia and Central West Asia is expected to deliver EIRRs higher than in Southeast Asia by 0.02% and 0.03%. Nevertheless, things go awry in Oceania, given that EIRR would reduce by 0.25%. The implication of this observation is limited since the sample size of Oceania's projects is small ($N = 2$). The independent variable Y2000 shows no statistical significance in all models, indicating that year of construction does not seem to be a factor (after controlling for general inflation).

Models 7 & 8 illustrate the correlation between *ex post* NPV and project features. As suggested by the negative coefficients for the independent variable implementation period, a longer implementation period severely diminishes projects' value. Longer roads are anticipated to generate higher NPV.

6. Discussion

This section first discusses three methodological issues associated with BCA for Asian Development Bank projects and then qualitatively reviewed the causes of the discrepancies between expected and actual project performances.

6.1. The choice of social discount rate

Weighing the upfront investment against the future social well-being is critical when engaging BCA to evaluate the worthiness of a public project. The accuracy and credibility of the evaluation results are largely determined by the appropriateness of the social discount rate (SDR) used. Applying a high SDR eliminates many projects which are less economic viable in the near future but may generate large inter-generational benefits in the distant future. In contrast, adopting a low SDR would expose the lenders to excessive risks and reduce their expected return on investment.

In line with the approach prescribed by other multilateral development organization specialising in supporting countries in under-developed regions, ADB follows a weighted average approach when determine the SDR, reconciling the opportunity cost of public funding in private investment or private consumption (Asian Development Bank, 2013). ADB adopts a uniform Social Discount Rate (SDR) of 12% in both *ex ante* analysis and *ex post* evaluation when calculating NPV, which ensures the comparability of BCA results in both stages (Asian Development Bank, 2016 b). This SDR is also regarded as the benchmark EIRR guiding project selection. ADB states that in *ex ante* stage, projects with an EIRR of higher than 12% or lower than 10% will be accepted or rejected respectively. For projects with an EIRR between 10% and 12%, decisions are made depending on the existence of provable extra economic benefits. In *ex post* evaluation, the recalculated project EIRR is compared with this SDR, which is a key determinant judging the

efficiency of project implementation. Although a range of 10%–12% has been designated, ADB encourages discretion in project SDR based on factors including geographical region, industry section, economic condition and project-specific features.

This 12% SDR appointed by ADB is far higher than that adopted by EU Commission's, which is 5%, and than that used by other developed countries like Canada (10%), France (4%), Italy (5%), New Zealand (10%) and the UK (3.5%). The differences in the applied SDRs among different regions across the world reflect the different marginal opportunity costs of public capital and projects' impact on inter-generational equity. Overall, the opportunity cost of public capital in less developed countries is higher than that in developing countries, leading to a higher SDR in the former. This can be explained by concerns that scarce resources and funding, less sophisticated financial systems, and high market volatility in developing countries require an extra rate of return to attract limited transnational capital. Asian Development Bank (2013) put forward that a reassessment of the suitability of the age-old 10–12% SDR is needed. In 2017, ADB decided to replace the old 12% SDR by a new 9% SDR (Asian Development Bank, 2017). This replacement took into account the growing income levels in ADB member developing countries, lower transnational loan costs than in the past decades, and ADB's increasing emphasis on environmental projects that typically have a long-run impact over the distant future. All the projects discussed in this paper uniformly applied a 12% SDR without any discretion, which ensures the comparability between *ex ante* and *ex post* evaluation.

However, whether this is the appropriate value, or if it is too high – leading to underinvestment in infrastructure – or too low – leading to overinvestment – is an open question. Fig. 1 visualizes *ex ante* (as per the y-axis) and *ex post* (as per the x-axis) project BCR. The size of each circle indicates the percentage of ADB funding out of the total project cost, and the color represents project NPV (in US\$ Million). Basically, both forecast and actual BCRs are above 1, indicating the accumulated present value of benefits exceeds that of costs. In addition, informed by the eight circles distributing to the right of the average *ex post* BCR reference line, the proportion of ADB's disbursement is higher for projects with smaller NPV since the six circles in light blue are bigger than the two circles in dark blue. This observation does not suggest a negative causal relationship between *ex post* project NPV and ADB funding proportion, but means that ADB is likely to be the major or even exclusive investor for projects in relative small funding scale.

Fig. 2 illustrates the distribution of *ex ante* (as per the vertical axis) and *ex post* (as per the horizontal axis) project EIRRs for 47 road projects.

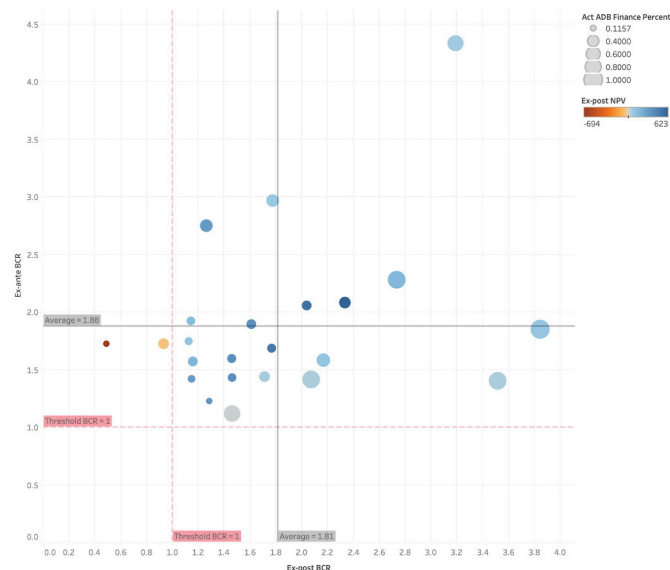


Fig. 1. Distribution of Asian Development Bank road projects' BCR (N = 23).

Each circle filled in different color represents a road project, with the size of the circle conveying the same information as Fig. 1. Given the 12% threshold EIRR stipulated by ADB, only one projects fall below this line in *ex ante* evaluation stage, and two projects failed to achieve an *ex post* EIRR exceeding this line. The average *ex ante* and *ex post* EIRR are 20.41% and 19.5% respectively, which are far higher than the 12% threshold.

In conclusion, the project outcomes informed by both BCRs and EIRRs demonstrate that ADB's practice conforms to or even exceeds the decision rules. The rigorous compliance secures return on investment to a large extent, where almost all projects generate BCR higher than 1 and EIRR higher than 12%. As a result, the adopted 12% discount rate at planning and post evaluation maintains the effectiveness of ADB's investment but at the same time indicates excessive risk aversion and consequent under-investment, as it is the portfolio of projects which should exceed a Benefit/Cost Ratio of 1, not every single project, and clearly they do so by a large margin. To avoid nearly every project falling below the threshold implies many projects especially those aiming to relieve regional poverty, were not funded.

6.2. The application of Non-Market Valuations

The monetary evaluation of transport projects' TTS benefits relies heavily on Non-Market Valuations (NMV). As non-marketable intangible benefits, TTS benefits are neither tradable nor priced by the market. In this case, shadow price is defined as the estimated unit price used to monetize TTS. In contrast to TTS benefits, VOC reductions can be gauged by considering factors like price of tires and auto parts, depreciation expense, and fuel price, which are available with market price. As a result, the risk caused by imprecise shadow price is negligible when evaluating VOC reductions.

Given that only by using fair and realistic shadow price can project benefits be accurately and credibly measured, our discussion about shadow price centers on two issues: 1) how shadow prices are determined and 2) whether shadow prices are properly adjusted. For the first issue, OECD countries generally consider revealed and stated preference of value of time for given origin-destination trips (Asian Development Bank, 2013). But it is much more difficult to acquire precise information in developing countries. According to ADB, journey time is evaluated separately for trips during working and non-working hours. For the former, ADB uses working wage as the fundamental element of shadow price, adds the overhead cost of employment like pension and holiday costs if applicable, and adjusts the ultimate shadow price with shadow



Fig. 2. Distribution of Asian Development Bank road projects' EIRR (N = 46).

wage conversion factor. For the latter, non-working time value is proportionate to local household income, especially when TTS only accounts for minor weight in terms of total project benefit.

However, engaging the user benefit assessment approach as the core benefit assessment method might largely overlook the wider benefits realised beyond the direct usage of the transport infrastructure. Because benefits accrued not only when people and goods can travel around faster (mobility) but also when reaching the destinations gains greater values (Levinson and King, 2019). Accessibility measures the ease of arriving at valuable destinations (COTAM, 2020; Hansen, 1959). The changes in land or real estate values resulting from the changes in accessibility are reflected by fluctuation in their market prices and hence provide an alternative path to assessing project benefits. The shadow price is largely determined by users' willingness to pay for transport, reflecting the perceived value of provisional travel decisions. Decisions regarding residential and commercial locations convey aggregate information and thus are more appropriate for evaluating both direct user benefits as well as indirect user and non-user benefits.

Furthermore, since project benefit evaluation covers multiple years after project delivery, shadow price is expected to be adjusted for changes in users' perception of time value which varies with productivity and income. In this case, the absence of a tangible market makes the NMV process more challenging. The valuation accuracy is subject to the accuracy of the base shadow price and the expected growth rate. An accessibility-based evaluation method, in contrast, is superior to the traditional NMV method due to the existence of the housing market and the availability to predict and verify the growth rate in a more credible manner. It is vital to notice that adding the NMV and accessibility-based methods leads to double counting. They are two distinct methods operating independently, but a comparison of their results may provide some practical insights.

For the second issue, the adjustment of shadow price is classified into adjustment to currency price and adjustment to variations in macroeconomic conditions. Both local price and world price are accepted in *ex ante* analysis. The distinction induced by using either local price or world price is pertinent whenever interventions in the trade lead to a disparity between those two types of prices. ADB measures and reports such a price disparity using the Standard Conversion Factor (SCF) and its inverse, the Shadow Exchange Rate Factor (SERF). Moreover, the assumptions based on which EIRR is calculated do not vary with change in price numeraire, thereby ensuring that the same EIRR is acquired using either price numeraire. Variations in the macroeconomic environment influence the users' perception of the value of time. Thus for developing countries with less stable economic environments, the adoption of two different sets of conversion factors in *ex ante* and *ex post* stage is justifiable.

6.3. The counterfactual base case

Project cost and benefits under various (at least one) "build" options are weighed against the corresponding "no-build" or "do-minimum" option, and the incremental approach captures the differences between them. The "no-build", as the name suggests, refers to a counterfactual circumstance where the prospective project does not exist. Given that the incremental approach is adopted in *ex ante* and *ex post* BCA, identifying proper counterfactual scenarios are required in both stages.

First, the consistency of the counterfactual scenario used in the two stages is of great concern. If the assessment of incremental cost and benefits hinges on distinct base scenarios, the output performance measurements are incomparable, thus negating the reliability of the conclusions on the accuracy of BCA. Considering the heterogeneity of projects conducted in different sectors and countries, ADB does not provide standardized guidance for determining the reference scenario, and project teams are encouraged to determine it at their discretion. After scrutinizing the Project Completion Reports (where the *ex post* and *ex ante* economic benefits are reported), we did not discover any further

specification about the choice of counterfactual case, presuming that no substantial amendments have been made on the initial choice. Although occasionally *ex ante* economic analysis were updated after the formal decision had been made, the *ex post* analysis is performed with respect to the updated *ex ante* version.

The second concern lies in the appropriateness of the counterfactual scenario used in both stages, particularly in the *ex post* stage. ADB states that the "no-build" base scenario may not necessarily be the current situation when preparing *ex ante* BCA. The extent to which the traffic facility is used currently (for projects aiming at replacing, rehabilitating, and modifying existing traffic facilities) and will be used in the foreseeable future in the absence of the envisaged action would form the reference base case. Routine maintenance costs in keeping reasonable road services are crucial when projecting cost streams for the "no-build" base case. In *ex ante* stage, the rationale of this setting is conceivable since the continuation and expansion of the actual demand observed in the absence of the future project are reasonable in contexts with growing demand. However, whether it remains to be an appropriate base case in the *ex post* stage is constrained by the unpredictability of how the external environment evolves as a result of the project and how the project responds to that evolution (Florio and Vignetti, 2013). To solve this problem, an observation period that is long enough to figure out how demand and supply, price, macroeconomic conditions, and demographic features actually evolve is essential to ensure a practical simulation of the potential evolution across decades after project completion in *ex ante* stage. The current practice in ADB where a PCR covers an actual observation time window of 1–2 years post to the project opening may be insufficient.

As described in section 3, the ADB currently carries out a two-step *ex post* project evaluation process where the first-step self-assessment needs to be undertaken by ADB's regional departments and the second-step independent assessment by ADB's IED. The two-step process could be carried on to succeeding *ex post* evaluation stages to maximize the advantages of a joint effort between the regional departments which could closely monitor project operation and the IED which provides independent justification and assurance.

6.4. Causes of the discrepancies in project performances

As informed by the results in section 5.2, the cost overrun of civil works ranks at the top in terms of both frequency and magnitude.

First, the unforeseeable price escalation of road construction materials, crude oil, and utility is a common cause of civil works cost overrun. The situation might worsen when the developed country mainly relies on importing those construction materials and its domestic currencies depreciate during project implementation. Further, experiencing high inflation during project implementation is also thought to increase civil costs, although we did not find statistical evidence for this hypothesis.

For instance, the civil work cost of Zamiin-Uud road in Mongolia was 2.5 times higher than that stated at appraisal (US\$ 50.06 million in appraisal and US\$ 135.93 million at completion) (Asian Development Bank, 2015 b). During the road construction process, the CPI of Mongolia rose from 100 in 2010 to 153.7 in 2014 (World Bank, 2020).

Second, partial or complete changes to the initial road design are likely to lift the cost of civil works. However, rather than technical design defects, the less regulated driving environment and illegal driving behaviours take some blame for the changes in project scope (Asian Development Bank, 2015 a).

Third, less than satisfactory performance of civil work contractors (both domestic and international contractors), like inefficient workers and material mobilization at the inception of project construction, being less proactive and responsive to project scope alteration, and withdrawing from the project in the middle of implementation, delay the overall project progress and incurs extra expenses. Park and Papadopoulou (2012) stated that granting civil work contracts to the lowest bidder runs high risks of over-budget, although we did not observe that

the lowest bidder was regarded as a risk factor in the projects covered by our study.

Last, since many of those projects have road rehabilitation components, worse than expected road conditions (after project approval) caused by environmental factors (faster than expected road erosion or collapse caused by spring thaw) or socioeconomic factors (social instability) requires additional investments to secure the successful completion of projects.

After civil works, land acquisition and resettlement rank next in terms of vulnerability to overspending. For road projects in urban areas, the expansion of the area of permanent land acquisition, the increases in the number of buildings to be demolished, the increased number of residents to be re-accommodated and compensated, as well as the increased compensation rates introduced by new government policies have been documented as causes of excessive costs. For instance, the Third Ring Road project in Xian, China, which links four major districts and two commercial zones in central Xian, incurred a land acquisition and resettlement cost of 3 billion in CNY, which doubled the budgeted amount (Asian Development Bank, 2011). For roads passing through arable fields where local resistance to road construction is strong, extra compensation is required for residents.

A common way of managing successful project delivery within budgets is reallocating the costs on less essential project components like the construction of local roads or the improvement to cross-border facilitation (many road projects have cross-boarder components) to the project components with higher priority like the construction of expressways, resulting in an average project construction costs overrun of 10.71%.

In terms of project benefits, an underestimation of both VOC and TTS is highlighted in section 5.2.

First, the traffic demands grow rapidly. The improved level of service and the shorter journey times, which are directly related to the project roads, facilitate growth. Such growth might be motivated by factors like the additional economic activities induced by the improved road networks. For example, a boom in contract farming in the vicinity of project roads in Lao People's Democratic Republic was identified when the project was about to complete, which turned out to facilitate the local farms to grow corn and other crops, followed by a sustainable increase in the export of those crops by using the project cross-border roads (Asian Development Bank, 2015 a). In contrast, some projects identified no generated traffic at the inception of project operation because either the data were unavailable or the increase in traffic demand is so substantial that they are likely driven by exogenous factors and thus accounted as normal traffic.

Second, the rise in Product Producer Index, labour costs fuel costs, tire costs and other costs that are of great importance in projecting VOC increase the *ex post* VOC savings.

Third, the actual growth rate of GDP per capita is higher than what was anticipated at the *ex ante* appraisal stage. GDP per capita is a crucial proxy used to predict the growth rate of traffic demand, unit travel time cost, and other unit prices used to project benefits.

7. Conclusion

Research into the performance of transport investments can be categorized into four groups: project costs, travel demand forecast modelling, other comprehensive project benefits, and overall project performance. Compared to the overall project performance, which is measured with indicators like NPV, EIRR or BCR, publications on the first three topics are relatively ample. The imbalanced research input for *ex post* overall project performance between developed countries (especially European countries) and developing countries is prominent, calling for empirical evidence of the latter.

This study covers 59 roadway projects funded by ADB. We scoured Project Completion Reports for all 59 projects to collect project information and assessed the accuracy of BCA results through four

dimensions: the frequency of error, the magnitude of error, the correlation of error and the tendency of error.

First, under an average project completion rate of 95.33%, we observed that the average construction cost overrun for 59 roadway projects is 10.71%, equivalent to USD 71.4 million. The cost of civil works and land acquisition rank at the top in terms of the likelihood and magnitude of suffering from *ex ante* underestimation. A 10% project contingency reserve was almost used up during construction and effectively functioned as a buffer mitigating the risk of cost overrun.

Second, grounded on 23 projects disclosing detailed economic analysis, we discovered a systematic tendency of understating both the present value of costs (18 out of 23 projects) and the present value of benefits (13 out of 23 projects) in *ex ante* BCA. As the two major components constituting transport project benefits, VOC savings and TTS are prone to be understated.

Third, more than half of projects (25 out of 47) underestimated EIRR and about 52.17% of them (12 out of 23) understated NPV. Nevertheless, the underestimation of economic benefits is too small to counterbalance the underestimation of costs, resulting in an overall project EIRR is 5.4% lower than the initial expectation. The vast majority of projects still have an *ex post* BCR above 1, suggesting under-investment or too strict a threshold for project funding.

Fourth, the two most statistically significant factors affecting total project cost are the proportion of funding injected by ADB and project length, and the direction of their correlations with project cost is opposite. Road type and project implementation period show significant impacts on percent cost error. A longer implementation period implies a higher percent error, and limited access roads generate cost overrun greater than roads containing limited access and access roads. Furthermore, the percent cost error shows clear geographical differences where projects in East Asia have a cost overrun greater than those in Southeast Asia.

Last, project type and construction sequence appear to be statistically significant estimators for project EIRR. In terms of geographic location, constructing 10 km of road in South Asia or Central West Asia is expected to deliver EIRR higher than in Southeast Asia.

We finally discussed ADB's choice of three key BCA elements: social discount rate, shadow price, and counterfactual base scenarios. First, applying a uniform 12% social discount rate in both *ex ante* and *ex post* stage ensures the accuracy and comparability of BCA results. The high discount rate (relative to the developed countries) successfully secures the overall return on investment, but indicates an excessive risk aversion and consequent under-investment by rejecting projects which would have produced an *ex post* BCR of 1. Second, the shadow price adopted when calculating TTS is determined fairly and adjusted realistically, thereby reinforcing the accuracy and credibility of BCA results. Whereas the NMV-based user benefit assessment approach might largely overlook the wider economic benefits realised beyond the direct usage of the transport infrastructure. In regard to the last question, we recommend that ADB either extend the observation period after project opening or prepare for subsequent *ex post* evaluation. This recommendation helps ascertain how demand and supply, price, macroeconomic conditions and demographic features actually evolve in *ex post* stage and thus ensures practical simulation of the potential evolution of project benefits and costs for decades after project completion in the *ex ante* project evaluation stage.

The causes of cost overruns include unforeseeable price escalation of construction materials, partial or complete changes to the initial road design due to the less regulated driving environment or illegal driving behaviours, less satisfactory performances of civil work contractors and worse than expected road conditions. The causes of benefit underestimations are the general growth of traffic demands, a higher-than-expected growth rate of GDP per capita and the rise in Product Producer Index, labour costs and other costs used to project VOC benefits. However, our discussions on those causes are qualitative. A systematic comparison with statistical evidence is required to explore further the

differences between the causes of cost overruns (and the underestimation of benefits) in developed and developing countries. Besides, an in-depth analysis of the accuracy of traffic demand forecasts is helpful to explain the reasons behind the underestimation of the benefits for road projects in developing countries.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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